CH1101: Quantum Chemistry and Spectroscopy

Tutorial Sheet 06

Module: Quantum Chemistry

Q1. Two valid one-electron wavefunctions are given below:

$$(i)\ \psi_{n,l,m_l}(r,\theta,\phi) = R_{3,0}(r)Y_{0,0}(\theta,\phi) \qquad (ii)\ \psi_{n,l,m_l}(r,\theta,\phi) = R_{3,2}(r)Y_{2,-1}(\theta,\phi)$$

What are the quantum numbers for each of the wavefunctions?

Q2. The ψ_{2s} atomic orbital for hydrogen-like atom is given as:

$$\psi_{2s} = \frac{1}{\sqrt{32\pi}}\left(\frac{Z}{a_0}\right)^{3/2}\left(2 - \frac{Zr}{a_0}\right)e^{-Zr/2a_0}$$

How many radial node do you expect to see for ψ_{2s}? Find, where the radial node of the wavefunction occurs in terms of a_0 for H, He^+, Li^{2+} atoms, and comment on your results.

Q3. A pair of $2p$ atomic orbitals are given as follows:

$$\psi_{2,1,\pm1} = \frac{1}{\sqrt{64\pi}}\left(\frac{Z}{a_0}\right)^{3/2}\left(\frac{Zr}{a_0}\right)e^{-Zr/2a_0}\ \sin\theta\ \mathrm{e}^{\pm \mathrm{i}\phi}$$

Show the following linear combinations are real:

$$(a)\quad \psi_{2p_x} = \frac{1}{\sqrt{2}}\left(\psi_{2,1,+1} + \psi_{2,1,-1}\right), \qquad (b)\quad \psi_{2p_y} = \frac{1}{\sqrt{2}\ i}\left(\psi_{2,1,+1} - \psi_{2,1,-1}\right)$$

Q4. The ψ_{1s} atomic orbital of a hydrogen is given as:

$$\psi_{1s} = \frac{2}{a_0^{3/2}}e^{-r/a_0}$$

Calculate at what distance from the nucleus the $1s$-electron of the hydrogen atom can be found with the maximum radial probability density, $r^2\ |\psi_{1s}|^2$.